

PROJECT 3: SPECIFICATIONS SHEET

DESCRIPTION OF THE PROJECT

You and your team have created a univariate time series classification algorithm, Rocket, and intend to send a short paper to a conference. The aim of the paper is to show that your algorithm outperforms the state-of-the-art and for this, you must design and execute an experimentation that will serve for this purpose and report the results and findings in a suitable manner. The conference you have chosen is one of the most prestigious conferences in the area of time series analysis, so it only has an acceptance rate of %25. Good luck!

TEAMS

The task will be carried out in groups of 3 students. Students can decide their group participants.

ASPECTS TO BE TAKEN INTO ACCOUNT IN THE EXPERIMENTATION

- **Code:** The algorithms can be programmed in any language of choice. A useful choice can be *aeon*, a toolkit for time series analysis in Python: <https://www.aeon-toolkit.org/en/stable/>.
- **Datasets:** The datasets to perform the experimentation can be obtained from the UCR/UEA time series classification repository: <https://timeseriesclassification.com/dataset.php>. These datasets can be downloaded and used directly in *aeon* using the *load_classification* function.

Choose a subset of the available datasets, taking into account that your choice must be fundamented in the paper. You must evaluate the algorithms in at least 5 datasets.

- **Benchmark algorithms:** Your algorithm must be compared with at least 6 benchmark algorithms with their default parameters: 1NN-Euclidean, 1NN-DTW, Shapelet Transform (ST), BOSS ensemble, Time Series Forest (TSF), InceptionTime and the Canonical Time-series Characteristics (catch22). You can include more benchmark algorithms and/or tune the hyperparameters if you want.
- **Validation framework:** Even if the datasets in the repository are already split into train/test splits, other validation schemes can be applied for more robust results (careful with computational time!).
- **Error metrics:** Accuracy should be used as a baseline metric, but other metrics can be used for a more profound analysis. Other aspects of the algorithms can also be evaluated, for example, computational time, robustness with respect to hyperparameters, etc.

REQUIREMENTS OF THE PAPER

- **Outline of the paper:** The paper should contain an abstract, an introduction, a section where your algorithm is explained, an experimentation section, a results section and some conclusions.

- **Format:** The paper should be written in Latex using the IEEE conference template.
- **Length:** 5 pages maximum (strict).

REQUIREMENTS OF THE ORAL PRESENTATION

- **Duration:** 10 minutes (strict) + questions
- **Outline:** Same as paper.

EVALUATION

It is required to participate and pass this task in order to be in continuous assessment, and the grade of the project will account for %20 of the final grade. Additionally, the best %25 of the presented papers will be selected and an extra 0.25 points will be added to the final grade of the course.

Merely fulfilling the minimum requirements will result in a 4/10.

Aspects that will be evaluated and will enhance your grade

- Suitability and complexity of the experimentation process: validation scheme used, metrics, choice of datasets, hyperparameter tuning, etc. (maximum 2 points)
 - Is the designed experimental process honest?
 - Is it suitable/complex enough to extract significant conclusions, given the available computational resources?
- Analysis of the obtained results: representation of the results and obtained conclusions. (maximum 1 point)
 - Are the results explored visually using graphical representation methods?
 - Have you tried to analyze the obtained results a bit more than just extracting the metrics?
 - Do you extract meaningful conclusions?
- Level of correctness and detail in the explanation of the algorithm in the paper and in the oral presentation. (maximum 2.5 points)
 - Have you understood the algorithm in detail?
 - Would you be able to reimplement it?
- Quality of the answers to the questions of the committee (0.5 points)

DEADLINE

The paper will be submitted on March 29th 2026 at 23:55 via eGELA.

The slides of the presentation will be submitted on March 26th 2026 at 11:00 via eGELA.

The oral presentation and feedback session will take place in class on the 26th of March.

USEFUL REFERENCES

Bagnall, A., Lines, J., Bostrom, A. *et al.* The great time series classification bake off: a review and experimental evaluation of recent algorithmic advances. *Data Min Knowl Disc* 31, 606–660 (2017). <https://doi.org/10.1007/s10618-016-0483-9>

Middlehurst, M., Schäfer, P. & Bagnall, A. Bake off redux: a review and experimental evaluation of recent time series classification algorithms. *Data Min Knowl Disc* 38, 1958–2031 (2024). <https://doi.org/10.1007/s10618-024-01022-1>

Aeon Python toolkit for time series mining. <https://www.aeon-toolkit.org/en/stable/>

Time Series Classification page. <https://www.timeseriesclassification.com/>

Dempster, A., Petitjean, F. & Webb, G.I. ROCKET: exceptionally fast and accurate time series classification using random convolutional kernels. *Data Min Knowl Disc* 34, 1454–1495 (2020). <https://link.springer.com/article/10.1007/s10618-020-00701-z>